1. Which are the top three variables in your model which contribute most towards the probability of a lead getting converted?

   Answer:

   Analyzing the coefficient values in the screenshot provided, the top three variables that contribute the most to the probability of a lead being converted are:
   1. Lead Source_Welingak Website
   2. Lead Source_Reference
   3. What is your current occupation_Working Professional

	coef
const	-0.0376
Do Not Email	-1.5218
Total Time Spent on Website	1.0954
Lead Origin_Landing Page Submission	-1.1940
Lead Source_Olark Chat	1.0819
Lead Source_Reference	3.3166
Lead Source_Welingak Website	5.8115
Last Activity_Olark Chat Conversation	-0.9613
Last Activity_Other_Activity	2.1751
Last Activity_SMS Sent	1.2942
Specialization_Others	-1.2025
What is your current occupation_Working Professional	2.6083
Last Notable Activity_Modified	-0.9004

2. What are the top 3 categorical/dummy variables in the model which should be focused the most on in order to increase the probability of lead conversion?

   Answer:

   Looking at the coefficient values in the screenshot provided, the top three categories/dummy variables to prioritize to increase the probability of lead conversion are:
   1. Last Activity_SMS Sent
   2. Lead Source_Welingak Website
   3. What is your current occupation_Working Professional

	coef
const	-0.0376
Do Not Email	-1.5218
Total Time Spent on Website	1.0954
Lead Origin_Landing Page Submission	-1.1940
Lead Source_Olark Chat	1.0819
Lead Source_Reference	3.3166
Lead Source_Welingak Website	5.8115
Last Activity_Olark Chat Conversation	-0.9613
Last Activity_Other_Activity	2.1751
Last Activity_SMS Sent	1.2942
Specialization_Others	-1.2025
What is your current occupation_Working Professional	2.6083
Last Notable Activity_Modified	-0.9004

3. X Education has a period of 2 months every year during which they hire some interns. The sales team, in particular, has around 10 interns allotted to them. So during this phase, they wish to make the lead conversion more aggressive. So they want almost all of the potential leads (i.e. the customers who have been predicted as 1 by the model) to be converted and hence, want to make phone calls to as much of such people as possible. Suggest a good strategy they should employ at this stage.

   Answer:

   In the provided image, the final prediction is determined using an optimal cutoff value of 0.37. To intensify sales efforts, the company can consider contacting all leads with a conversion probability (value = 1) below a cutoff of 0.3(highlighted in yellow).

	Converted	Converted_prob	Prospect ID	predicted	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	final_predicted	Lead_Score
0	0	0.196697	3009	0	1	1	0	0	0	0	0	0	0	0	0	20
1	0	0.125746	1012	0	1	1	0	0	0	0	0	0	0	0	0	13
2	0	0.323477	9226	0	1	1	1	1	0	0	0	0	0	0	0	32
3	1	0.865617	4750	1	1	1	1	1	1	1	1	1	1	0	1	87
4	1	0.797752	7987	1	1	1	1	1	1	1	1	1	0	0	1	80
5	1	0.744001	1281	1	1	1	1	1	1	1	1	1	0	0	1	74
6	0	0.100027	2880	0	1	1	0	0	0	0	0	0	0	0	0	10
7	1	0.965845	4971	1	1	1	1	1	1	1	1	1	1	1	1	97
8	1	0.854512	7536	1	1	1	1	1	1	1	1	1	1	0	1	85
9	0	0.768071	1248	1	1	1	1	1	1	1	1	1	0	0	1	77
10	0	0.594758	1429	1	1	1	1	1	1	1	0	0	0	0	1	59
11	0	0.056000	2178	0	1	0	0	0	0	0	0	0	0	0	0	6
12	0	0.056145	8554	0	1	0	0	0	0	0	0	0	0	0	0	6
13	1	0.540563	5044	1	1	1	1	1	1	1	0	0	0	0	1	54
14	1	0.904502	3475	1	1	1	1	1	1	1	1	1	1	1	1	90
15	1	0.429049	7424	0	1	1	1	1	1	0	0	0	0	0	1	43
16	0	0.144193	421	0	1	1	0	0	0	0	0	0	0	0	0	14
17	0	0.128530	3591	0	1	1	0	0	0	0	0	0	0	0	0	13
18	0	0.015142	6247	0	1	0	0	0	0	0	0	0	0	0	0	2
19	0	0.357168	7843	0	1	1	1	1	0	0	0	0	0	0	1	36

4. Similarly, at times, the company reaches its target for a quarter before the deadline. During this time, the company wants the sales team to focus on some new work as well. So during this time, the company's aim is to not make phone calls unless it's extremely necessary, i.e. they want to minimize the rate of useless phone calls. Suggest a strategy they should employ at this stage.

   Answer:

   To reduce the number of unproductive phone calls, the company can consider contacting all leads with a conversion probability (highlighted in yellow) under column 0.7. However, it's important to note that there is a possibility of missing out on some leads that are actually converted but have been inaccurately predicted as not converted (highlighted in red in the image below). This is not a significant concern as the overall target has already been achieved.

	Converted	Converted_prob	Prospect ID	predicted	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	final_predicted	Lead_Score
0	0	0.196697	3009	0	1	1	0	0	0	0	0	0	0	0	0	20
1	0	0.125746	1012	0	1	1	0	0	0	0	0	0	0	0	0	13
2	0	0.323477	9226	0	1	1	1	1	0	0	0	0	0	0	0	32
3	1	0.865617	4750	1	1	1	1	1	1	1	1	1	1	0	1	87
4	1	0.797752	7987	1	1	1	1	1	1	1	1	1	0	0	1	80
5	1	0.744001	1281	1	1	1	1	1	1	1	1	1	0	0	1	74
6	0	0.100027	2880	0	1	1	0	0	0	0	0	0	0	0	0	10
7	1	0.965845	4971	1	1	1	1	1	1	1	1	1	1	1	1	97
8	1	0.854512	7536	1	1	1	1	1	1	1	1	1	1	0	1	85
9	0	0.768071	1248	1	1	1	1	1	1	1	1	1	0	0	1	77
10	0	0.594758	1429	1	1	1	1	1	1	1	0	0	0	0	1	59
11	0	0.056000	2178	0	1	0	0	0	0	0	0	0	0	0	0	6
12	0	0.056145	8554	0	1	0	0	0	0	0	0	0	0	0	0	6
13	1	0.540563	5044	1	1	1	1	1	1	1	0	0	0	0	1	54
14	1	0.904502	3475	1	1	1	1	1	1	1	1	1	1	1	1	90
15	1	0.429049	7424	0	1	1	1	1	1	0	0	0	0	0	1	43
16	0	0.144193	421	0	1	1	0	0	0	0	0	0	0	0	0	14
17	0	0.128530	3591	0	1	1	0	0	0	0	0	0	0	0	0	13
18	0	0.015142	6247	0	1	0	0	0	0	0	0	0	0	0	0	2
19	0	0.357168	7843	0	1	1	1	1	0	0	0	0	0	0	1	36